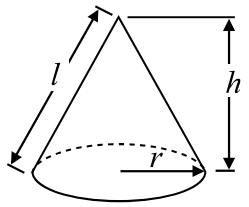
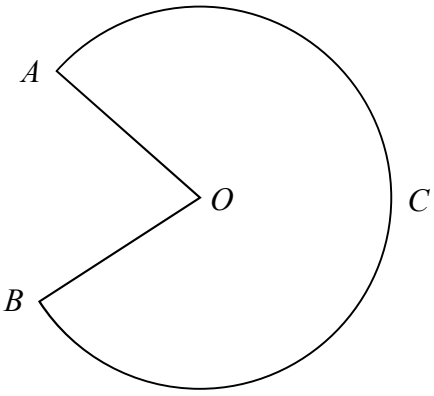
23 The diagram shows a sector $OACB$ of a circle with centre O.
The point C is the midpoint of the arc AB.

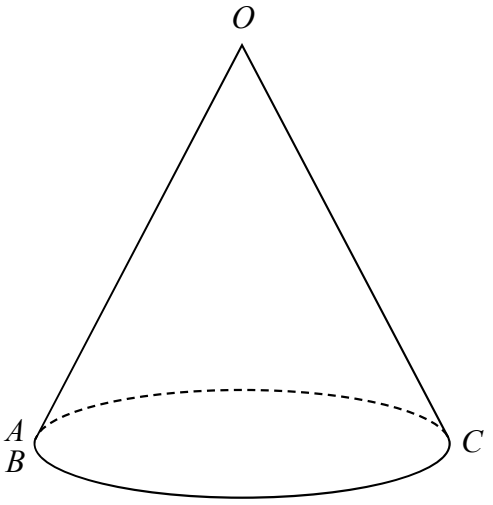
The diagram also shows a hollow cone with vertex O.
The cone is formed by joining OA and OB.

Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$







The cone has volume $56.8\,\text{cm}^3$ and height $3.6\,\text{cm}$.

Calculate the size of angle AOB of sector $OACB$.
Give your answer correct to 3 significant figures.
You must show all your working.

$56.8 = (1/3)\pi r^2 \times 3.6$, so $r^2 = 56.8/(1.2\pi)$.

Base radius $r \approx 3.88158$ cm.

Slant radius $l = \sqrt{(r^2+3.6^2)} \approx 5.29402$ cm.

Sector radius is l; its arc is the cone base rim.

$(\theta/360) \times 2\pi l = 2\pi r$, so $\theta = 360r/l$.

$\theta \approx 263.952° \approx 264°$ (3 s.f.).

5 marks total
Cone volume
Find base radius
Pythagoras for l
Match arc lengths
264°

°

(Total for Question 23 is 5 marks)